Atkia Mahila

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Linkedin — Github

Experience

- **Graduate Research Assistant** *Aug 2025 – Present*
  Rochester Institute of Technology,
  Department of Computing and Information Sciences *Rochester, NY*
  - Led experimental evaluation of tree-style LLM inference on vLLM and SGLang, quantifying how batching and cache behavior shape end-to-end latency and GPU efficiency.
  - Developed lightweight tracing and analysis to separate prefill- vs. decode-dominated regimes and pinpoint when recomputation becomes the primary cost driver.
  - Conducted research in high-performance and distributed computing, focusing on scalable reproducibility analytics and checkpoint comparison for large-scale simulations.
  - Contributed to a CPU-side, asynchronous checkpoint-analysis workflow for early detection of reproducibility deviations, emphasizing throughput-oriented I/O handling and hierarchical hashing.
- **Senior Associate, IT Core Applications** *Mar 2024 – Jul 2025*
  Robi Axiata PLC *Bangladesh*
  - Supported and improved a production Loyalty Management System under reliability constraints and time-sensitive operational requests.
  - Automated recurring workflows (interconnect billing and operational tasks) using Python,Java and shell scripting, reducing manual steps and repeated errors.
  - Maintained containerized deployments, performed server health checks, and improved data workflows using SQL.
- **System Engineer** *Sep 2023 – Mar 2024*
  Huawei Technologies (Bangladesh) *Bangladesh*
  - Supported telecom charging and billing operations using scripting and PLSQL for routine checks and issue resolution.
  - Contributed to system stability and security by addressing operational incidents and mitigating vulnerabilities.
  - Supported large rollout work (Teletalk CBS Expansion Project, 2023–2024).
- **Industrial Trainee** *Fall 2022*
  BJIT (Bangladesh Japan IT Limited) *Bangladesh*
  - Completed an industry training program with hands-on software engineering project work.

Technical Skills

Languages: Python, C++, Bash
ML / LLM: PyTorch, HuggingFace, vLLM, SGLang
Systems: Linux, Docker, profiling/monitoring , structured logging
Data / Infra: SQL, PL/SQL, metrics/logging pipelines, scripting/automation

Projects

- **Tree-style LLM generation performance study (vLLM / SGLang / HuggingFace)**
  - Instrumented branching/tree-style generation runs and collected throughput, latency, end-to-end time, GPU memory, cache hit rate, batching stats, token lengths, prefill time, and prefill-compute tokens using structured traces.
  - Analyzed preemption behavior and its performance impact, including how scheduling and cache/memory pressure can trigger additional prefill/recompute and shift latency.
  - Investigated shallow-tree tendencies in practice (limited depth), relating them to runtime constraints such as token budgets, batching dynamics, and observed cache/memory pressure signals.
- **Early detection of reproducibility deviations via checkpoint analysis and offset-aware random-read I/O**
  - A study on CPU-based runtime for online, error-bounded checkpoint comparison that detects reproducibility deviations early during execution without interfering with GPU compute.
  - Implemented Merkle-tree hierarchical hashing and I/O batching to improve comparison throughput; evaluated on HACC-style checkpoint workloads with up to 7× speedup over baseline methods.
  - Studied I/O patterns dominated by small, non-contiguous reads and applied offset coalescing plus a regression-based model to select batching/coalescing parameters that balance I/O and computation throughput.

Education

- **Ph.D., Computing and Information Sciences** — *Aug 2025 – Present*
  Rochester Institute of Technology — *Rochester, NY*
- **B.S., Computer Science and Engineering** — *Jan 2018 – Aug 2023*
  Chittagong University of Engineering and Technology (CUET) — *Bangladesh*
  CGPA: *3.43/4* (72.60%) — Class Position: 34th out of 126
  Last four semester's GPA: *3.70/4*